

ENGR 2303 Electrical Science
Review Material and Sample Test for Test 2

Given Electric circuit diagram, you should be able to find the current, voltage, and power at any element in the circuit using any of the following techniques:

1. Superposition
2. Thevenin's Theorem
3. Norton's Theorem
4. Operational Amplifier

Question 1

Find I_o in the circuit in **Fig. 1** using superposition.

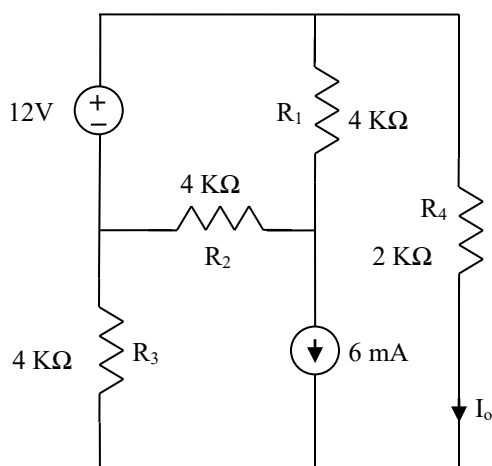


Fig. 1

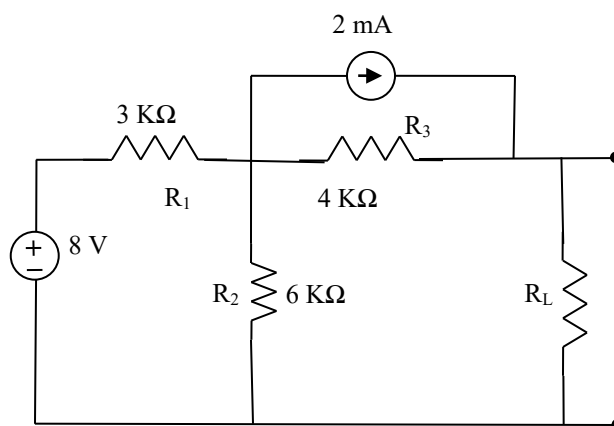


Fig. 2

Question 2

Find I_o in the circuit in **Fig. 1** using Norton's Theorem.

Question 3

Find R_L for maximum power transfer and the maximum power that can be transferred to the load in **Fig. 2** above.

Question 4

Derive the gain of the operation amplifier $G = V_o/V_{in}$ in terms of the resistors only. Show your derivation for credits.

